

Math 421

Friday, October 3

Announcements

- Homework 4 due **today**
 - Double check that you have correctly indicated on which pages each problem is answered!
- Exam 1 – 10/15
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!

Divergence

Def: A sequence that does not converge is said to diverge.

Examples:

Discussion Questions

- ❖ Write the negation of the definition of convergence.
- ❖ What proof method might you use to show a sequence diverges?
- ❖ Discuss how you might show that $(-1)^n$ diverges.

Example

Theorem. The sequence $a_n = (-1)^n$ does not converge.

Proof.

Limits are Unique

Thm: If $\lim s_n = s$ and $\lim s_n = t$, then $s = t$.

Q: How many different ways can you think of to prove two quantities are equal?

Limits are Unique

Thm: If $\lim s_n = s$ and $\lim s_n = t$, then $s = t$.

Proof.

Chapter 2: Sequences

Section 9: Limit Theorems for Sequences

Bounded

Definition. A sequence (s_n) is said to be **bounded** if the set $\{s_n : n \in \mathbb{N}\}$ is a bounded set.

Activity –

1. Give an example of a bounded and an unbounded sequence.
2. Does every bounded sequence converge? Does every unbounded sequence diverge?

Theorem 9.1

Theorem. *Convergent sequences are bounded.*

Discussion.

Proof.

Algebraic Limit Laws

Theorems 9.2-9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then,

- i. $\lim ks_n = ks$ for all $k \in \mathbb{R}$.
- ii. $\lim s_n + t_n = s + t$.
- iii. $\lim s_n t_n = st$.
- iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$
- v. if $t \neq 0$, then $\lim \frac{s_n}{t_n} = \frac{s}{t}$.

FALSE Proof of ii. $\lim s_n + t_n = s + t$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $N > 0$ such that if $n > N$, then $|s_n - s| < \frac{\epsilon}{2}$. Since $\lim t_n = t$, there exists $N > 0$ such that if $n > N$, then $|t_n - t| < \frac{\epsilon}{2}$. Assume $n > N$. Then

$$\begin{aligned} |(s_n + t_n) - (s + t)| &= |(s_n - s) + (t_n - t)| \leq |s_n - s| + |t_n - t| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n + t_n = s + t$.

FALSE Proof of iii. $\lim s_n t_n = st$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists $N_1 > 0$ such that if $n > N_1$, then $|s_n - s| < \frac{\epsilon}{2(|t|+1)}$. Since $\lim t_n = t$, there exists $N_2 > 0$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon}{2|s_n|}$. Let $N = \max\{N_1, N_2\}$.

Assume $n > N$. Then

$$\begin{aligned} |s_n t_n - st| &= |s_n t_n - s_n t + s_n t - st| \leq |s_n t_n - s_n t| + |s_n t - st| \\ &\leq |s_n| |t_n - t| + |t| |s_n - s| < |s_n| \frac{\epsilon}{2|s_n|} + |t| \frac{\epsilon}{2(|t|+1)} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n t_n = st$.

FALSE Proof of v. if $t \neq 0$, then $\lim \frac{s_n}{t_n} = \frac{s}{t}$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since (t_n) is convergent, there is a constant $M > 0$ such that $|t_n| \leq M$ for all n . Since $\lim s_n = s$, there exists $N_1 > 0$ such that if $n > N_1$, then $|s_n - s| < \frac{\epsilon M}{2}$. Since $\lim t_n = t$, there exists $N_2 > 0$ such that if $n > N_2$, then $|t_n - t| < \frac{\epsilon |t| M}{2|s|}$. Let $N = \max\{N_1, N_2\}$. Assume $n > N$. Then

$$\begin{aligned} \left| \frac{s_n}{t_n} - \frac{s}{t} \right| &= \left| \frac{s_n t - s t_n}{t_n t} \right| = \left| \frac{s_n t - s t + s t - s t_n}{t_n t} \right| \leq \left| \frac{s_n t - s t}{t_n t} \right| + \left| \frac{s t - s t_n}{t_n t} \right| \\ &= \left| \frac{s_n - s}{t_n} \right| + \left| \frac{s(t - t_n)}{t_n t} \right| < \frac{1}{M} \frac{\epsilon M}{2} + \frac{|s|}{M|t|} \frac{\epsilon |t| M}{2|s|} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{s_n}{t_n} = \frac{s}{t}$.

Diverging to $\pm\infty$

Definition 9.8

- $\lim s_n = +\infty$ provided for each $M > 0$ there is a number N such that $n > N$ implies $s_n > M$.
- $\lim s_n = -\infty$ provided for each $M < 0$ there is a number N such that $n > N$ implies $s_n < M$.

Example: $\lim n = +\infty$.

Terminology: A sequence (s_n) has a **limit** or the **limit exists** if (s_n) converges to a real number or diverges to $\pm\infty$.

Activity

1. Prove, directly from the definition, that $\lim(\sqrt{n} + 7) = +\infty$.
2. Prove that the unbounded sequence $s_n = (-1)^n n$ does not have $-\infty$ or $+\infty$ as a limit.
 - a) What does it mean for an unbounded sequence to not diverge to $+\infty$ or $-\infty$, i.e. what is the negation of the definition?
 - b) What proof technique should you use?

Caution!

The limit laws we've previously stated do not apply to sequences that have infinite limits, we need to prove these separately.